

# Report: Network Security & Scanning

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**Program:** Cybersecurity & Ethical Hacking

## 1. Objective

The goal of this task is to perform active reconnaissance on the target system (**Metasploitable2**). I will identify open ports, discover running services, and analyze network traffic to understand the attack surface of the target.

## 2. Scanning Environment Verification

Before scanning, I verified that the lab environment was active and isolated using the **Host-Only** network configuration established in Task 1.

- **Attacker IP (Kali Linux):** 192.168.56.101
- **Target IP (Metasploitable2):** 192.168.56.102
- **Connectivity Test:** I performed a ping command from Kali to Metasploitable2 to ensure the path was clear for scanning.

## 3. Service and Version Discovery (Nmap)

This is the first "live" step. We need to know which "doors" are open and what's behind them.

**Action:** Run the following command in your Kali terminal:

```
nmap -sV -T4 192.168.56.102
```

**Why this command?**

- **-sV:** This probes open ports to determine **service/version info**. (Crucial for finding exploits later!)
- **-T4:** This sets the timing template to "Aggressive," making the scan faster for a lab environment.

## 4. Operating System Fingerprinting

Identifying the OS is a vital step in reconnaissance because exploits are usually OS-specific.

**Action:** Run the following command in Kali:

```
Bash
sudo nmap -O 192.168.56.102
```

- **The Logic:** Nmap sends "probes" (TCP and UDP packets) to the target and analyzes the response. Every OS (Linux, Windows, etc.) has a slightly different way of responding—like a digital fingerprint.
- **Findings:** Your report should state that the target was identified as a **Linux 2.6.x** kernel, which confirms it is an older, vulnerable system perfect for our lab testing.

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### 5. Identifying High-Risk Vulnerabilities (The "Low Hanging Fruit")

Based on your Nmap service scan from the previous step, we will now highlight the most dangerous findings for your mentors at **ApexPlanet**.

| Port | Service | Version       | Why it is High Risk                                                                         |
|------|---------|---------------|---------------------------------------------------------------------------------------------|
| 21   | FTP     | vsftpd 2.3.4  | This version contains a famous <b>backdoor</b> that allows instant root access.             |
| 23   | Telnet  | Linux telnetd | Telnet sends all data (including passwords) in <b>Plain Text</b> , making it easy to sniff. |
| 445  | Samba   | 3.x - 4.x     | Often vulnerable to "EternalBlue" or "SambaCry" exploits.                                   |

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### 6. Network Traffic Analysis (The "Evidence")

- **Step:** Open **Wireshark** while running your Nmap scan.
- **Observation:** In your report, mention that you observed a high volume of **TCP SYN** packets.
- **The "Why":** This proves you understand the **TCP Three-Way Handshake** and how scanners use "half-open" connections to find ports without completing a full connection.

### 7. Automated Vulnerability Assessment (Nessus)

Automated scanners compare the services you found in the previous step against a massive database of "Known Vulnerabilities."

- **Process:**
  1. Launch the Nessus service in Kali.
  2. Start a "Basic Network Scan."
  3. Enter your **Metasploitable2 IP** as the target.
- **Observations:** In your report, you should note how the scanner categorizes risks into **Critical, High, Medium, and Low**.
- **Findings:** You will likely see a "Critical" alert for **VNC Password 'password'**. This indicates that the system is using a default, easily guessable password.